Amateur Radio & Basic RF Electronics Study Notes

This study guide introduces beginner concepts in amateur radio, antennas, RF propagation, frequency, wavelength, and polarization.

1. Ohm’s Law

Voltage, current, and resistance are related by Ohm’s Law: $V = IR$. This is one of the most important formulas in electronics.

2. Frequency and Wavelength

Radio frequency and wavelength are inversely proportional. The relationship is: wavelength = speed of light / frequency.

3. Example Calculation

If wavelength = 30 meters, frequency = 300,000,000 / 30 = 10 MHz.

4. HF, VHF, and UHF

HF supports long-distance communication through ionosphere reflection. VHF and UHF are mostly line-of-sight communication systems.

5. Antenna Height

Increasing antenna height improves communication range, especially for VHF and UHF systems. Higher antennas reduce obstacles and increase radio horizon distance.

6. Vertical Polarization

A vertically mounted antenna creates vertical polarization. Common in mobile radios, repeaters, and handheld communication.

7. Horizontal Polarization

A horizontally mounted antenna creates horizontal polarization. Often used for HF amateur radio communication.

8. Human Body and RF

The human body can absorb and affect RF energy, especially at VHF/UHF frequencies where wavelengths are closer to body size.

9. Quarter-Wave Antenna

Many antennas are designed using fractions of wavelength. Quarter-wave antennas are very common in amateur radio.

10. Security Concepts in Software Projects

Enterprise software projects use authentication, session management, password hashing, rate limiting, and dependency scanning tools such as Black Duck and SonarQube.

Band	Frequency Range	Typical Use
HF	3–30 MHz	Long-distance communication
VHF	30–300 MHz	Line-of-sight, repeaters
UHF	300 MHz–3 GHz	Wi-Fi, mobile systems

Prepared for beginner learning in amateur radio, RF engineering, and secure software development concepts.